

Types and Components of Computer System



Learning Objectives

Explain the basic parts of a computer system

Identify the uses and components of internal and external hardware

Hardware and Software

Hardware

- It refers to the physical components of a computer system that you can touch and see.

Software

- It refers to the non-tangible components of a computer system, including programs, applications, and operating systems, that instruct the hardware on how to perform specific tasks and functions.

LET'S DISCUSS HARDWARE

Two Categories of Hardware

External

- It refers to physical components of a computer system that are not housed inside the main unit.
- These components typically connect to the computer externally via ports or cables.

Internal

- It refers to the physical components of a computer system that are installed inside the computer's case.

LET'S DISCUSS EXTERNAL
HARDWARE

2 Categories of External Hardware

Input devices

- Used to send data to the computer.
- keyboards, mice, scanners, camera, microphones, touchscreens, joysticks, barcode and scanners.

Output devices

- Used to deliver information back to you.
- monitor, printer, speaker, headphones, projector, and plotter.

LET'S DISCUSS INTERNAL
HARDWARE

Examples of Internal Hardware

1

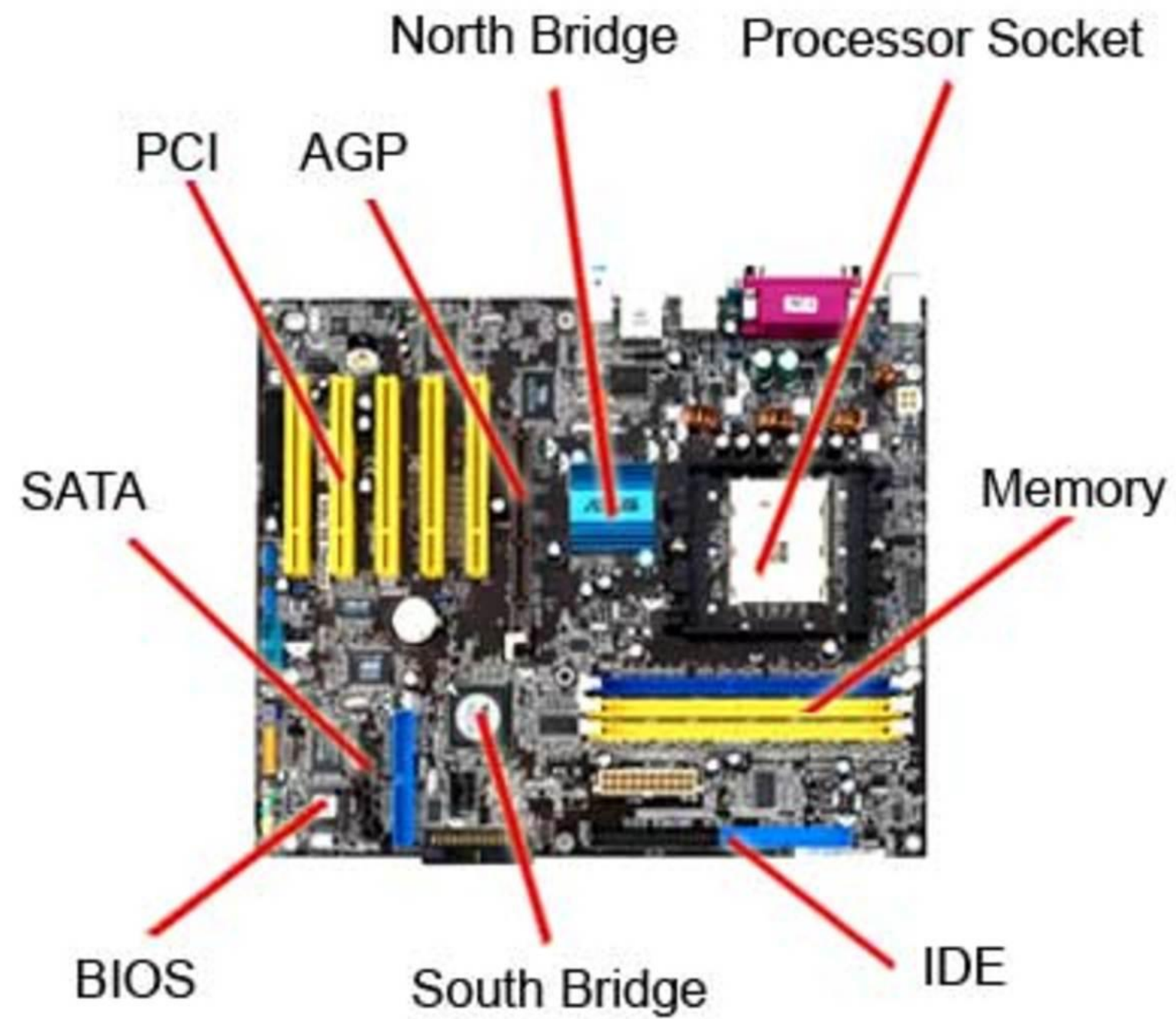
Motherboard

- It is the main circuit board inside a computer that holds and connects all the essential components.
- It ensures that these parts can work together to make your computer run.

- It also has ports and slots for connecting external devices like your keyboard, mouse, and monitor.

1

Motherboard



Examples of Internal Hardware

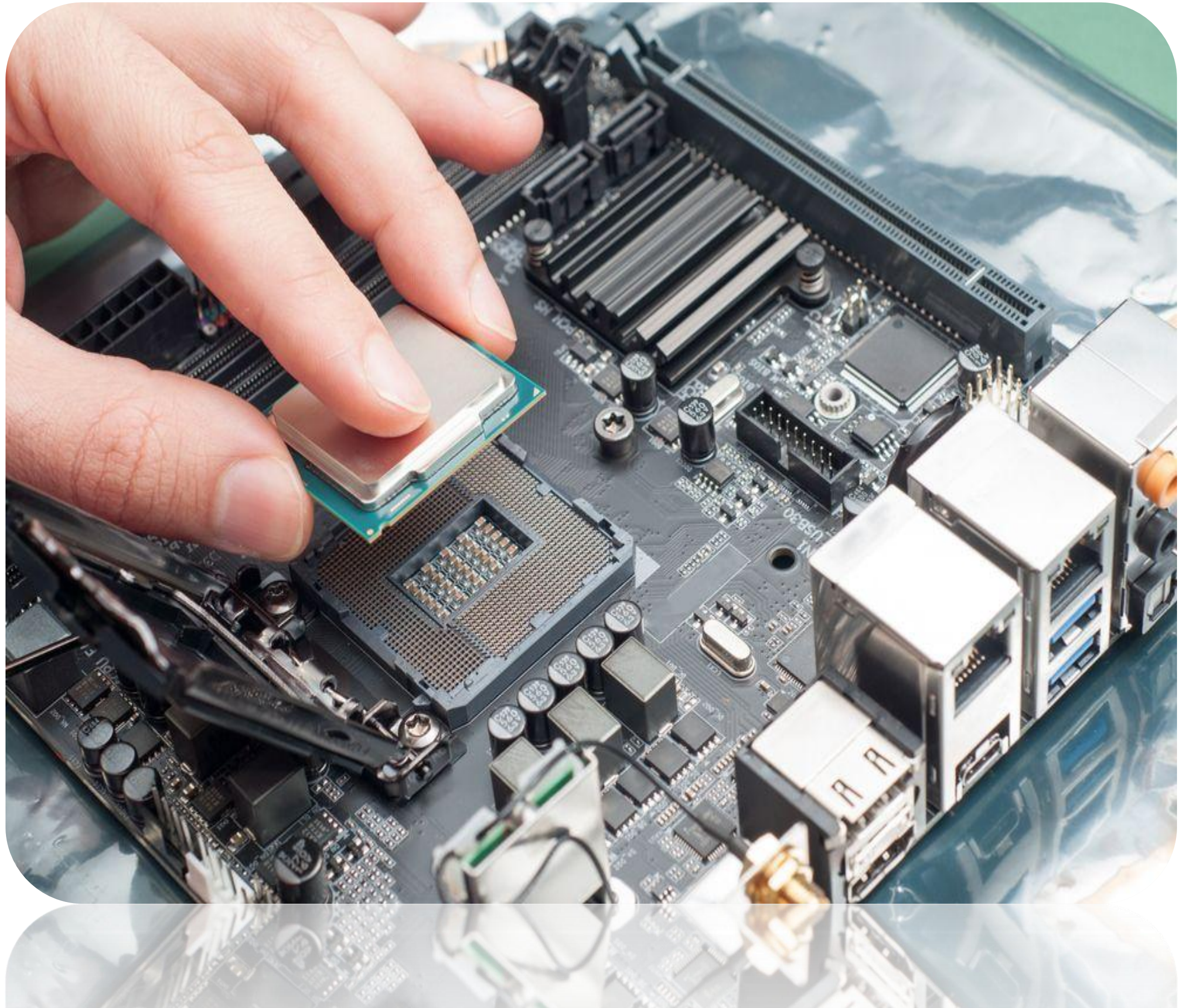
2

CPU

- **Central Processing Unit.**
- It's like the ***brain of a computer.***
- It's a tiny but super important chip that does all the thinking and calculations necessary to make your computer work. When you click on something or run a program, the CPU processes that information and carries out the instructions.

2

CPU



Examples of Internal Hardware

3

RAM

- **Random Access Memory**
- It's like your computer's short-term memory.
- It doesn't permanently store data, it only holds temporarily while the computer is powered on. When you turn off your computer, everything in RAM get erased.

3

RAM



Examples of Internal Hardware

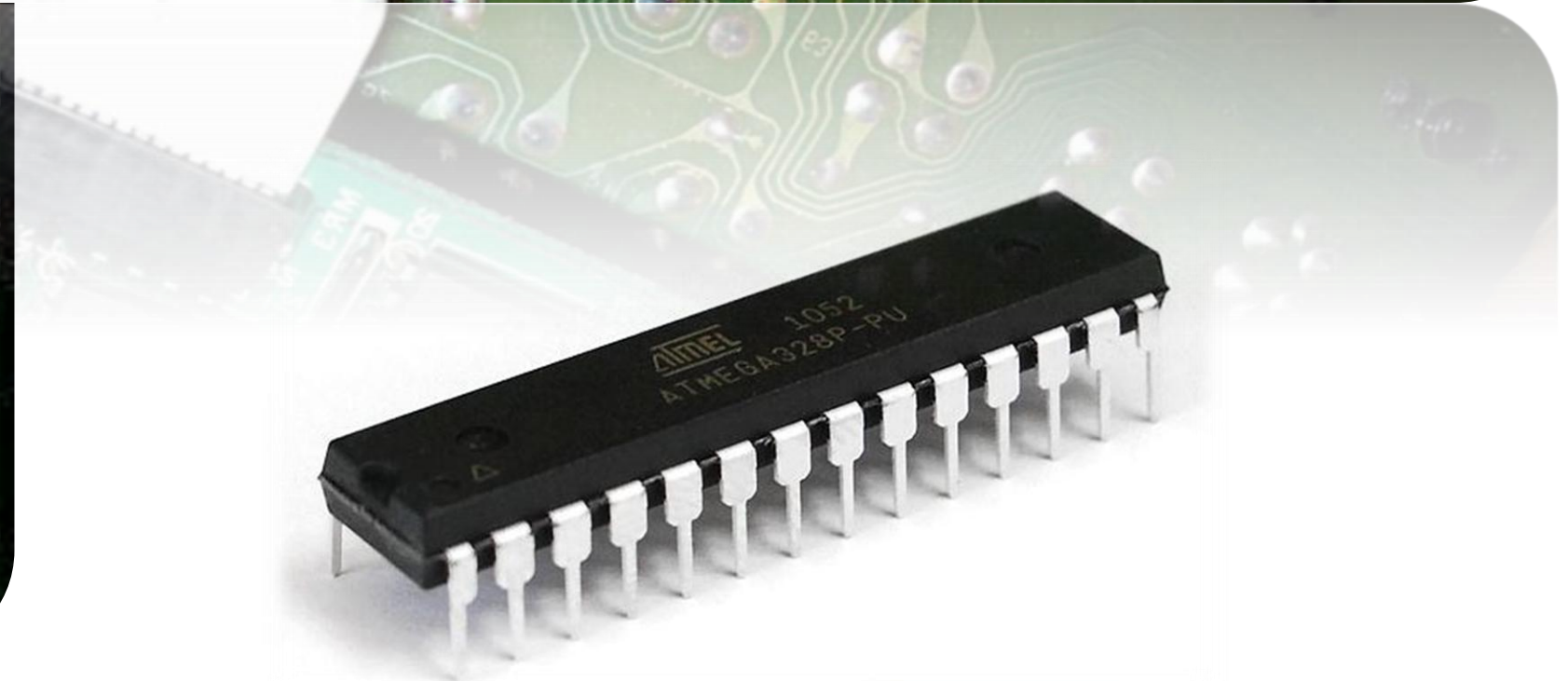
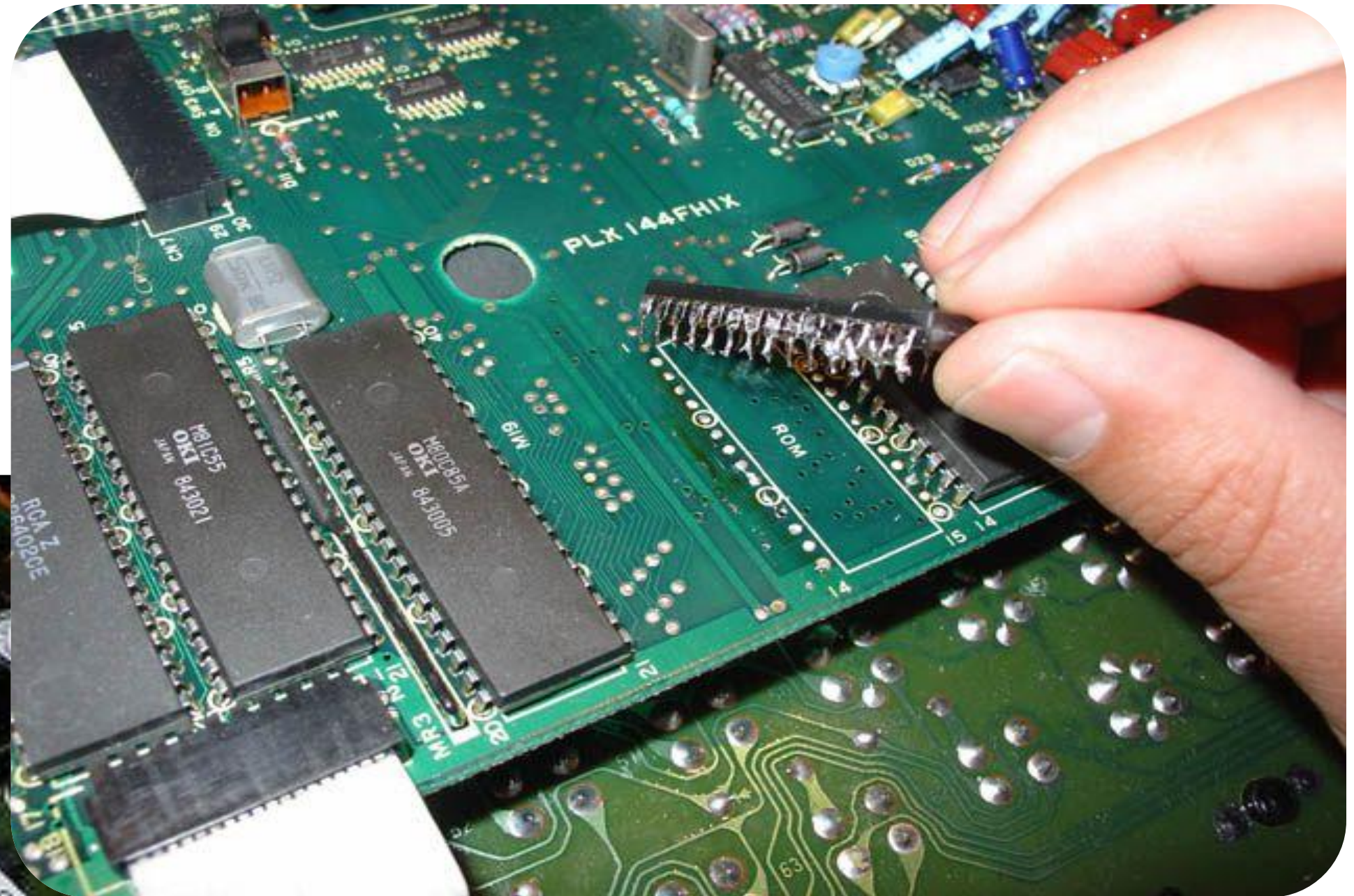
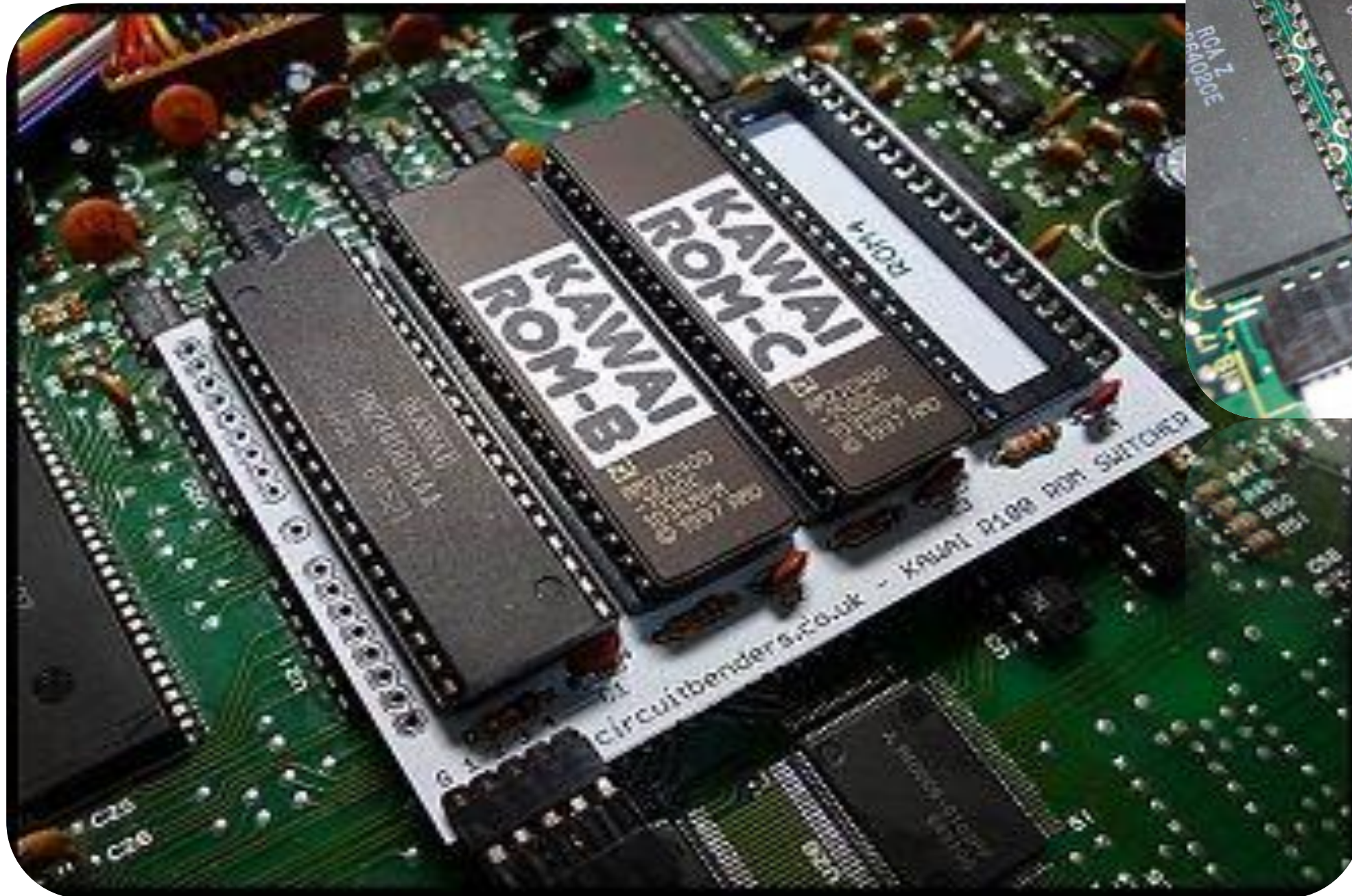
4

ROM

- **Read-Only Memory**
- It's like your computer's long-term memory.
- It is designed to store permanent instructions and data that remain intact even when the computer is powered off unlike **RAM**.

4

ROM



Examples of Internal Hardware

5

GRAPHICS CARD

- also known as a GPU (**Graphics Processing Unit**)
- It's like a supercharged artist inside your computer.
- Its main job is to handle all the visual stuff you see on your screen, like images, videos, and animations.

5



Examples of Internal Hardware

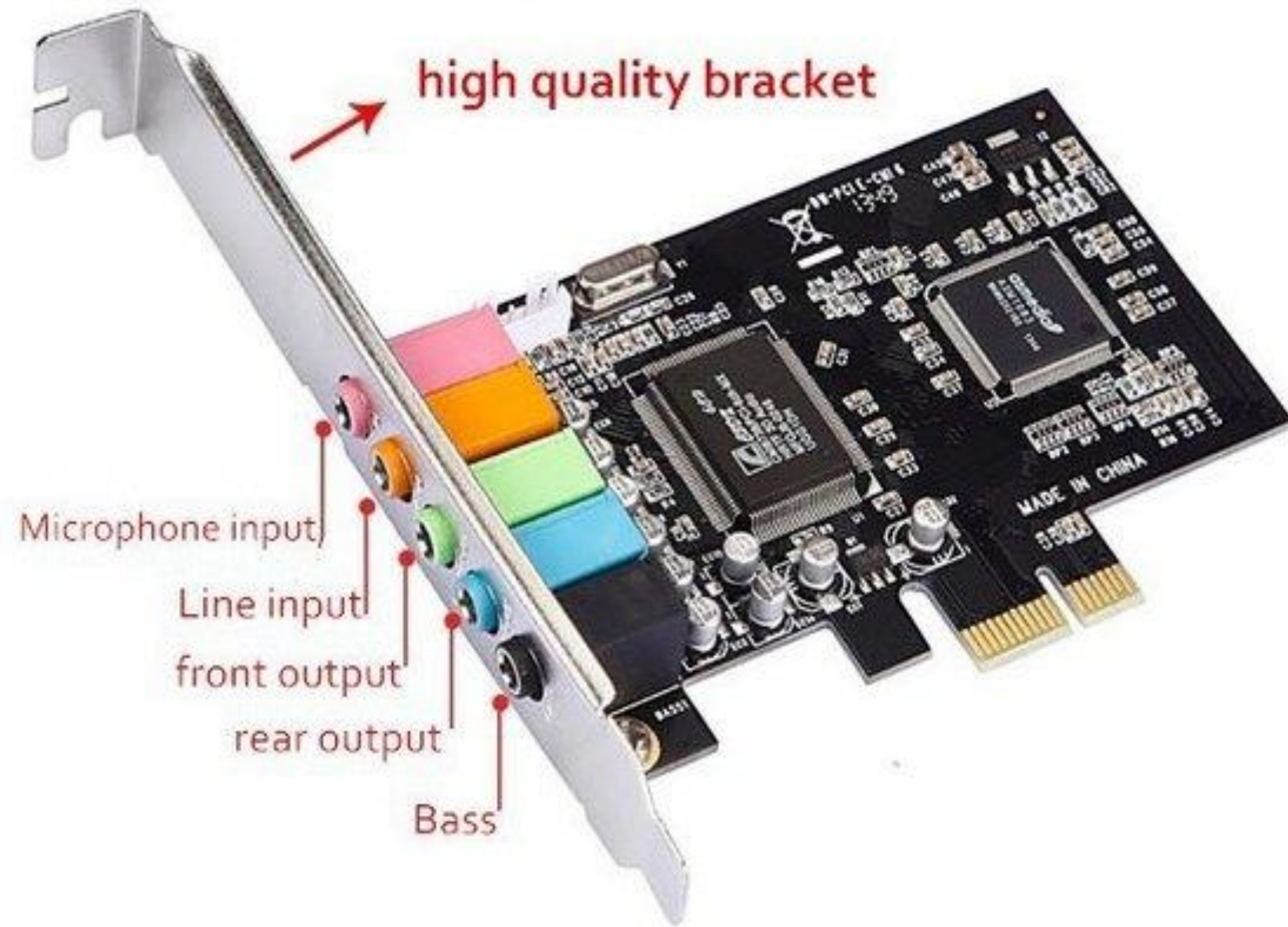
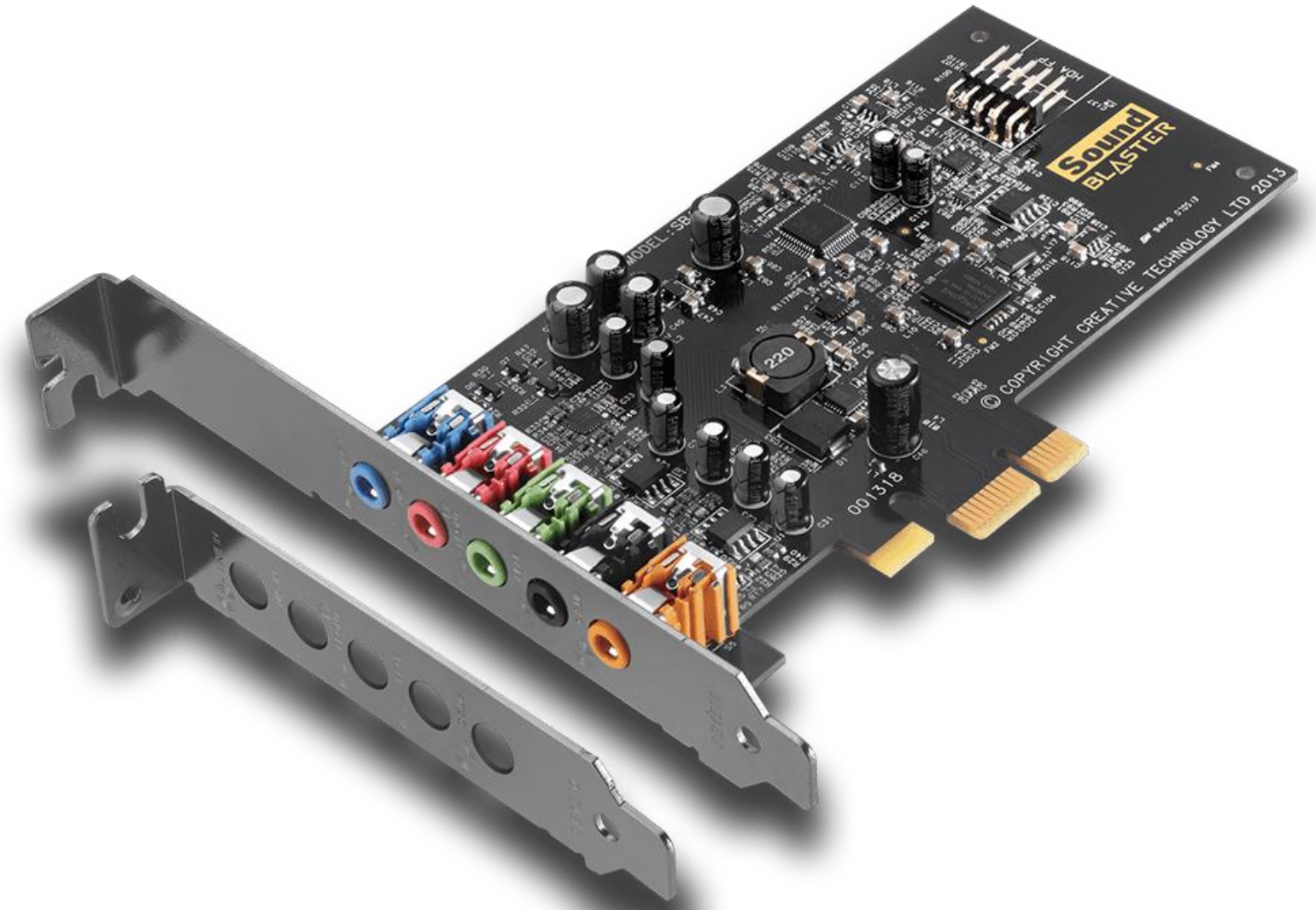
6

SOUND CARD

- It's like a conductor for your computer's audio orchestra.
- It's responsible for producing and processing sound signals, allowing you to hear music, videos, games, and other audio on your computer.

6

SOUND CARD



Examples of Internal Hardware

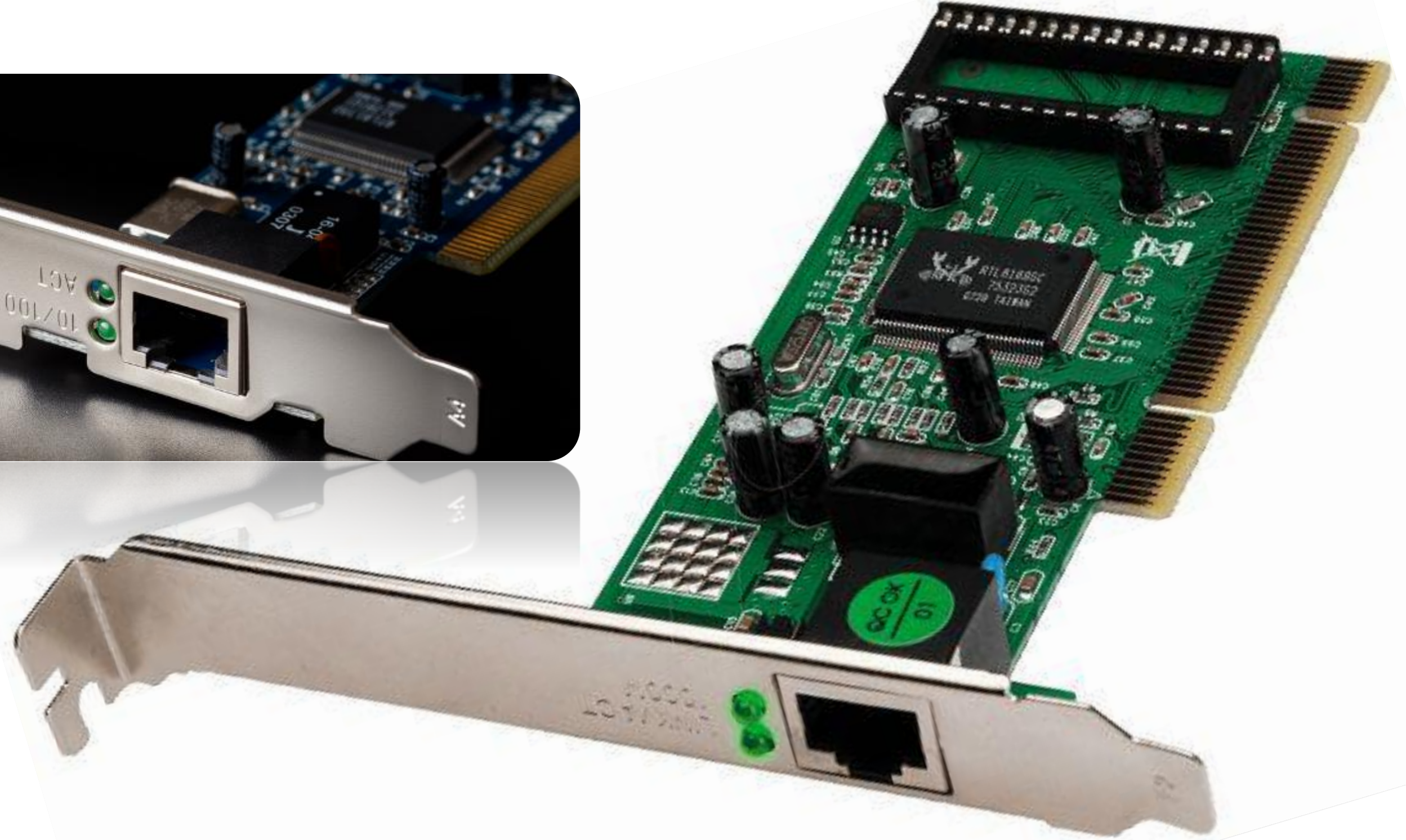
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NIC

- **Network Interface Card**
- also known as a network adapter
- is like a messenger that helps your computer communicate with other devices on a network.
- It's a hardware component that connects your computer to a network, whether it's a local area network (LAN) in your home or office, or the vast network of the internet.

7

Network Interface Card



Examples of Internal Hardware

8

INTERNAL STORAGE DEVICES

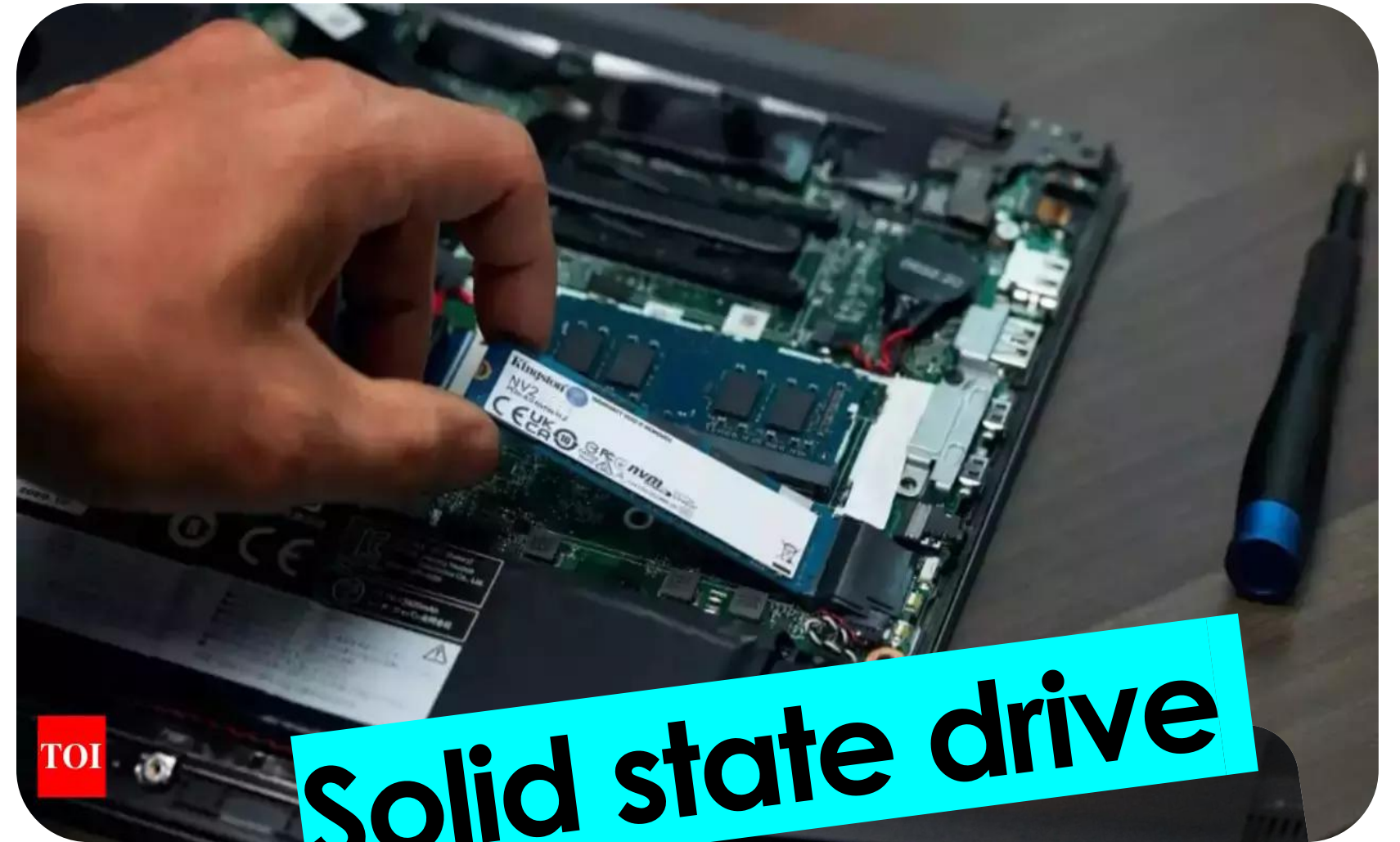
- Hard disk drive and solid-state drive
- are like the closets and drawers inside your computer where it keeps all its important stuff.
- It stores data, programs, and operating systems permanently, so they're always available whenever you need them.

8

INTERNAL STORAGE DEVICES



Hard disk drive



Solid state drive



Examples of Internal Hardware

9

POWER SUPPLY

- Part inside the computer that changes the electricity from the wall into the right type of power the computer parts need to work. It sends power to important components like the motherboard, CPU, and hard drives. Without a power supply, the computer cannot turn on or run. It also helps protect the computer by making sure the power is steady and safe.

8

POWER SUPPLY



Power Supply

Examples of Internal Hardware

10

COMPUTER FAN

- It is a small device inside the computer that helps keep it cool by blowing air over the parts. It stops the computer from getting too hot, which can cause damage or make it slow down. The fan makes sure the computer works well and lasts longer by keeping the temperature at a safe level.

10

COMPUTER FAN



Computer Fan

ACTIVITY TIME